

Project Title
(Optimal Binary Search Tree Designer)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Optimal Binary Search Tree (OBST) Generator and Search Cost Analyzer is a Java-based DSA project that creates a Binary Search Tree with the lowest possible search cost. The system takes sorted keys and their search frequencies as input. Since some keys may be searched more often than others, the project finds the best arrangement of the keys so that frequently searched keys can be found faster.

The project uses Dynamic Programming, especially interval Dynamic Programming, to find the best arrangement. It calculates the search cost for different groups of keys and selects the best key as the root of each group. The calculated values are stored in DP tables and reused to avoid repeated calculations.

Finally, the system constructs and displays the Optimal Binary Search Tree and shows the minimum search cost, expected search cost, and DP table. The project demonstrates important DSA concepts such as Binary Search Trees, Dynamic Programming, recursion, and search-cost optimization.

Signature of the Guide